David Geddam

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Portfolio, LinkedIn, GitHub

Education

Bob Jones University — Greenville, SC
Bachelor of Science in Computer Science — *Expected graduation: May 2027*

Papers

Fault prediction in Wind Turbines via Joint-Embedding Predictive Architecture — *Jan. 2026*
Developing · [Abstract] [Code]

- Building a Self-Supervised World Model that learns latent dynamics of 52-dim. sensor data from 95 wind turbines across 3 wind farms, grouping sensors into Controls, Environment, and State subsystems.
- Implementing VICReg loss to prevent representation collapse during self-supervised training, with Transformer-based encoder learning causal relationships between physical subsystems.

World Model verifiers for out-of-distribution policy safety — *Jan. 2026*
Developing · [Paper] [Code]

- Current RL agents and LLM-driven robots are "black boxes"; they don't have a world model to tell them they're about to violate a physical boundary.
- Proposing a world model that quantifies latent surprise via non-reconstructive predictive architectures to verify out-of-distribution safety violations.

Experience

Lovable — New York City Metropolitan Area
Campus Leader — *01/2026 – present*

- Selected to represent AI software development platform, managing partnerships, and organizing AI-assisted development hackathons.

Independent Research — New York City Metropolitan Area
Research Assistant — *12/2025 – present*

- Supervisor: Sumanth Ratna. Kandavalli (ratna@nyu.edu).

Projects

Tiny Siri - Edge-Optimized Intent Classification | *Python, PyTorch, Transformers, Streamlit* — November 2025

- Built a voice intent classifier by fine-tuning DistilBERT with a custom data augmentation pipeline, achieving 97 percent test accuracy and deploying it via Hugging Face Spaces and Streamlit. Optimized the model for on-device deployment using PyTorch dynamic quantization, reducing memory footprint by 48 percent (255MB → 132MB) while maintaining full precision.

Medical diagnosis with Hybrid Vision Transformers | *PyTorch, CNNs, Transformers* — October 2025

- Built a deep learning pipeline using a hybrid MobileNetV2 + Transformer architecture with self-attention mechanisms to classify brain dementia types and lung cancer malignancies, achieving 98 percent and 95 percent test accuracy respectively. Also used the Transformer's spatial feature processing to identify localized malignant patterns, with Grad-CAM heatmaps providing explainability.

Adversarial Vision Attacks | *Python, PyTorch* — September 2025

- This project demonstrates the fragility of modern Deep Learning Computer Vision systems (specifically ResNet-50) against Adversarial Attacks. By exploiting the gradients of the model, I successfully "poisoned" images that look identical to humans but are completely misclassified by the AI.

Skills

Programming: Python, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Docker, Git, CI/CD, Cursor